Closing the Arabic→English Gap in Multilingual Retrieval: Cross-lingual Fine-tuning, Distillation, and an MTEB Arabic Benchmark

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Abstract

Multilingual embedding models are strong when query and passage share a language and much weaker when an Arabic query must find an English passage. On a held-out benchmark built from public question-answering data, the 568M-parameter bge-m3 reaches MRR@10 63.3 for ar→en and 67.8 for en→ar. We fine-tune it with LLM-translated cross-lingual queries, mined hard negatives and a cached contrastive loss, then interpolate the weights back towards the original model (WiSE-FT). The result, v2-m3, scores 73.5 (ar→en) and 87.3 (en→ar), and its macro average over four directions rises from 74.8 to 84.4. Distilling it into 278M and 118M students lifts their ar→en score to 64.5 and 60.2 (off-the-shelf multilingual-E5-small: 41.4), though they stay well below the teacher. We also report what this costs: relative to bge-m3, v2-m3 changes by −3.3 points on MIRACL (Arabic) and −7.3 on Mr.TyDi re-ranking, and we show that 408 of the 1,000 MIRACL Arabic dev queries are near-verbatim questions from the TyDi QA training set. A 19-task Arabic slice of MTEB used for these checks, and int8 CPU deployment numbers, are included.

1 Introduction

Retrieval-augmented systems for Arabic users routinely search documents written in both Arabic and English. A multilingual bi-encoder is supposed to make this a non-issue, yet the published multilingual models are markedly asymmetric: in our benchmark the same model that scores 87.8 for English-to-English retrieval scores 63.3 when the query is Arabic and the passage English, and 67.8 in the opposite direction. A first version of this work (v1) fine-tuned a 118M multilingual-E5-small and lifted en→ar from 61.7 to 80.4 (above bge-m3's 67.8), but left ar→en at 55.1, 8.2 points behind bge-m3.

This report describes v2 and is deliberately specific about what did and did not work. Our contributions are: (i) a larger, contamination-filtered cross-lingual training set (20,999 new translated queries) with a documented licence trail; (ii) three models: **v2-m3** (fine-tuned bge-m3, weight-interpolated), and two distilled students, **v2-base** (278M) and **v2-small** (118M); (iii) a guard-task evaluation that reports the forgetting we observe on MIRACL and Mr.TyDi instead of hiding it, with a contamination analysis of those tasks; (iv) MTEB(ara, v1), a 19-task Arabic benchmark we assembled for these checks; and (v) measured CPU deployment costs for ONNX and dynamic int8 variants.

2 Related Work

Benchmarks. MTEB (Muennighoff et al., 2023) and its massively multilingual successor MMTEB (Enevoldsen et al., 2025) define the task suites in which our Arabic benchmark is expressed. Arabic retrieval is covered by MIRACL (Zhang et al., 2023), Mr.TyDi (Zhang et al., 2021), built on TyDi QA (Clark et al., 2020), and by the parallel QA sets XQuAD (Artetxe et al., 2020) and MLQA (Lewis et al., 2020). **Models.** Multilingual E5 (Wang et al., 2024, 2022) and M3-Embedding (bge-m3) (Chen et al., 2024) are the strong open multilingual encoders we start from; the latter is trained with self-knowledge distillation. **Training.** We use in-batch contrastive training (Henderson et al., 2017; Reimers and Gurevych, 2019) with gradient caching to reach large batches on one GPU (Gao et al., 2021), mined hard negatives and the false-negative problem they bring (Xiong et al., 2020; Qu et al., 2021), and embedding-level distillation (Hinton et al., 2015; Hofstätter et al., 2020). **Preserving the base model.** Interpolating fine-tuned and original weights, WiSE-FT (Wortsman et al., 2021), trades in-domain gain for robustness; we apply it to a retrieval embedder and measure the

Training queries	built	dropped	kept
Human questions (TyDi, ARCD, SQuAD)	16,419	1,173	15,246
Translated, v1	4,499	254	4,245
Translated, new in v2	20,999	1,113	19,886
Total (v2 recipe)	41,917	2,540	39,377

Table 1: Training queries of the v2 recipe before and after the contamination filter.

trade on held-out tasks.

3 Data

Benchmark. The benchmark of v1 uses only public data: TyDi QA GoldP (Apache-2.0), ARCD (MIT), SQuAD v1.1 (Rajpurkar et al., 2016; Mozannar et al., 2019) (CC BY-SA 4.0) and XQuAD (CC BY-SA 4.0), split into article-disjoint train, dev and test pools. The test split has 6,823 queries over 9,817 passages; XQuAD's parallel questions give four directions (ar→ar, en→en, ar→en, en→ar) with identical information needs. Each slice is searched in the passages of its own pool in the passage language, and we report MRR@10 per direction and the macro average of the four directions.

Translated queries. For cross-lingual training pairs, *questions only* are translated by a locally served LLM (Gemma-4-26B-A4B) into the other language; the passage keeps its language and stays the positive. v1 used 4,499 such pairs; v2 adds 20,999, taken from questions whose passages v1 never paired across languages. Test queries are never machine-translated. As a style check, translated XQuAD ar→en queries score 5.1–6.2 points higher than the human questions on the same 1,190 items for every model we tried, a near-constant offset rather than evidence of an over-fit translation style.

Contamination filter. While auditing the evaluation sets we found that 408 of the 1,000 Arabic dev queries of MIRACL-HardNegatives (41%) are near-verbatim questions (token Jaccard ≥ 0.8) of the TyDi QA Arabic *training* set, which v1 trained on. Measured against the full TyDi-ar and MIRACL-ar training questions the number is 816, leaving 184 clean queries. v2 therefore drops every training question that is a near-duplicate of a query of any audited task (7 MTEB tasks), and every training question whose positive passage overlaps an evaluation positive. After filtering, the number of near-duplicate MIRACL queries is 0. Table 1 gives the counts.

4 Method

Objective and negatives. All models minimise an InfoNCE loss over in-batch passages plus one mined hard negative per query, taken from ranks 1–30 of a baseline retriever in the same language pool, keeping only candidates at least 0.02 cosine below the positive to limit false negatives. v2-m3 uses a cached multiple-negatives loss (Gao et al., 2021) with batch size 64, mini-batch 16, similarity scale 50, learning rate 1e-5, one epoch, and maximum length 256 tokens.

Distillation. The e5-base and e5-small students use the standard loss (scale 20, lr 3e-5) plus a KL term between the student's and the frozen teacher's softmax over the in-batch scores (temperature 20, weight 1); the teacher is the raw fine-tuned bge-m3 (before WiSE blending).

WiSE interpolation. The fine-tuned bge-m3 weights are blended with the original ones, $\theta = \alpha\,\theta_{\text{FT}} + (1-\alpha)\,\theta_{\text{bge-m3}}$. We fixed the selection rule before any guard-task evaluation: the blend must stay within 0.5 dev macro points of the raw model and within 1 point of bge-m3 on a mono-Arabic mMARCO proxy (Bonifacio et al., 2021); this selected $\alpha = 0.75$.

5 Results

Setup. Model selection used the DEV split only; TEST was scored for final candidates. Confidence intervals are percentile bootstraps over queries (2,000 resamples, within slices), paired when comparing systems. All fine-tuned numbers are single seeds unless stated; Table 6 lists what is still running.

5.1 Main results

Table 2 shows the test results and Table 3 the paired differences. Fine-tuning bge-m3 on the cross-lingual data and blending the weights back (v2-m3) lifts ar→en by +10.1 points and en→ar by +19.6; the macro average moves from 74.8 to 84.4. Distillation transfers part of that gain: with the same data, a student trained without the teacher scores 60.7 (base) and 58.0 (small) on ar→en, and with it 64.5 and 60.2. Both students remain far below the teacher's 73.9. The 278M v2-base is statistically indistinguishable from the off-the-shelf 568M bge-m3 on ar→en (the paired interval includes zero) and ahead of it on the macro average, at about half the parameters.

Model	ar→ar	en→en	ar→en	en→ar	macro	95% CI (macro)
mE5-small	76.8	87.8	41.4	61.7	66.9	[66.0, 67.9]
v1 (e5-small FT)	80.2	87.3	55.1	80.4	75.7	[74.9, 76.6]
e5-small FT, no KD	79.4	86.4	58.0	80.2	76.0	[75.2, 76.9]
v2-small (KD)	80.8	87.6	60.2	81.1	77.4	[76.6, 78.3]
e5-base FT, no KD	78.9	86.3	60.7	81.4	76.8	[76.0, 77.7]
v2-base (KD)	81.8	88.6	64.5	84.1	79.8	[78.9, 80.6]
bge-m3	80.2	87.8	63.3	67.8	74.8	[73.9, 75.7]
v2-m3 raw FT	**87.1**	90.0	**73.9**	**88.4**	**84.8**	[84.1, 85.6]
v2-m3 (WiSE 0.75)	86.7	**90.0**	73.5	87.3	84.4	[83.6, 85.1]
mE5-base (not fine-tuned)	[TODO: baseline not yet scored on the TEST split]					

Table 2: Test MRR@10 $\times 100$ per direction. Models are grouped by size (118M, 278M, 568M). Bold: best of the column. v1 is seed 13 (the three-seed mean of v1 is lower in ar→en); "no KD" rows use the same data as the distilled model.

	Δ ar→en		Δ macro	
Comparison (B vs A)	pts	95% CI	pts	95% CI
v1 vs mE5-small	+13.7	[+11.8, +15.6]	+8.8	[+8.1, +9.6]
v2-small vs v1	+5.1	[+3.9, +6.3]	+1.7	[+1.3, +2.1]
KD effect, small	+2.2	[+1.3, +3.1]	+1.4	[+1.1, +1.7]
KD effect, base	+3.8	[+2.6, +5.0]	+2.9	[+2.5, +3.3]
v2-m3 vs bge-m3	+10.1	[+8.7, +11.6]	+9.6	[+9.0, +10.3]
WiSE cost (raw to 0.75)	−0.4	[−1.0, +0.2]	−0.4	[−0.7, −0.2]
v2-base vs bge-m3	+1.2	[−0.9, +3.1]	+5.0	[+4.2, +5.8]

Table 3: Paired differences on the test split (percentage points).

DEV selection set	ar→en	macro	held-out MIRACL-ar
e5-base student (teacher: raw FT bge-m3)			
no KD	71.0	78.8	–
KD w=1	74.5	81.2	–
KD w=3	75.6	81.8	73.1
KD only	76.1	82.0	73.2
bge-m3 fine-tuned itself			
bge-m3 (no FT)	73.5	78.2	80.5
raw FT	81.0	85.3	79.1
WiSE α=0.75	80.5	84.9	79.8
WiSE α=0.50	79.8	84.2	79.8
bge-m3 + MIRACL-ar replay / preservation			
replay	80.8	84.9	78.7
preservation	80.9	84.9	79.0
preservation, WiSE 0.75	80.6	84.7	79.8

Table 4: DEV MRR@10 $\times 100$ for the ablations; the last column is a held-out slice of MIRACL-ar training questions used only for selection.

	ar→en		macro	
Student	partial	full	partial	full
e5-small	56.7	58.0	76.2	76.0
e5-base	59.9	60.7	77.3	76.8
bge-m3	72.6	73.9	84.2	84.8

Table 5: Test MRR@10 $\times 100$ with the translated queries available at the first runs (partial) versus the full v2 set.

5.2 Ablations

Table 4 collects DEV results. For the students, a larger distillation weight and KL-only training were at least as good on dev as weight 1, so the weight used here is conservative; the final student recipe may change (pending). Re-weighting towards MIRACL (replaying MIRACL-ar training questions in the mix) and a feature-preservation loss to the frozen bge-m3 on Arabic positives did not improve a held-out MIRACL-ar training slice beyond what plain WiSE gave (bge-m3 itself: 80.5), so v2-m3 keeps the simpler recipe. Table 5 shows that more translated data helped bge-m3 but did not help the two small students on test. Across three seeds the raw v2-m3 spans 1.1 points on ar→en (s.d. 0.6) and 0.6 on macro (Table 6).

5.3 MTEB(ara, v1)

To check that gains on our benchmark do not come at the expense of general Arabic ability, we assembled MTEB(ara, v1): 19 MTEB tasks (retrieval, re-ranking, STS, pair classification, classification, clustering, bitext mining) with Arabic data, scored for 10 models (Table 7). bge-m3 leads at 72.0 mean over tasks. The v1 fine-tune (65.0) ends *below* its own starting point multilingual-E5-small: the cross-lingual gain was paid for on tasks that do not look like our training data, in particular Mr.TyDi re-ranking (79.2 to 65.8) and MIRACL (72.4 to 65.8). [TODO: final MTEB(ara) scores of v2-small, v2-base and v2-m3 are still being computed]

5.4 Guard tasks and the contamination analysis

Table 8 reports three tasks we treat as guards, in the MTEB protocol. Relative to bge-m3, v2-m3 is +2.7 points on SemRel24, but −3.3 on MIRACL-

TEST	ar→en	macro
v2-m3 raw FT, seed 13	73.9	84.8
v2-m3 raw FT, seed 14	73.2	84.3
v2-m3 raw FT, seed 15	72.7	84.3
mean ± s.d. (3 seeds)	73.3 ± 0.6	84.5 ± 0.3
v2-m3 WiSE 0.75, seeds 14 and 15	[TODO: blends pending]	
v2-base / v2-small, seeds 14 and 15	[TODO: runs pending]	

Table 6: Seed variation on test; pending runs are marked.

HardNegatives (75.7 vs 79.0) and −7.3 on Mr.TyDi re-ranking. Our pre-registered bar was to stay within 1 point of bge-m3 on MIRACL while keeping test ar→en at 72 or above; it is *not met*, and neither replay nor preservation closed the gap.

Contamination does not explain the gap away. Splitting the MIRACL and Mr.TyDi queries into those that are near-duplicates of TyDi-ar or MIRACL-ar training questions and those that are not (Table 9), the difference between v2-m3 and bge-m3 on the clean MIRACL queries is −4.4 points (all queries: −5.4). Only 184 MIRACL queries are clean, so the interval is wide, and the split was computed with a slightly different scoring input than the MTEB protocol. [TODO: split scores are 3.4 pts below the MTEB protocol for bge-m3 (titles missing); rerun pending] The reverse caveat also holds: bge-m3 lists MIRACL and Mr.TyDi among its training sets, so its absolute MIRACL number is not a clean reference either.

6 Deployment

We export v2-small and v2-base to ONNX (fp32, opset 17, pooling and normalisation inside the graph) and compare three ways of shrinking them for CPU serving: ONNX Runtime dynamic int8 with per-tensor or per-channel weight scales (activations quantised on the fly, cf. Zafrir et al., 2019), and weight-only 8-bit block quantisation of the MatMul weights that leaves the activations in fp32. Quality is measured as MRR@10 on the same test split against the PyTorch fp32 model, on CPU, with paired bootstrap intervals. Latency was measured on the 20-core ARM CPU of a shared server (not an x86 laptop or cloud VM), one process, 2 and 8 threads; numbers are therefore indicative.

Table 10 shows that the ONNX fp32 graphs reproduce the PyTorch results (+0.00 and +0.00 macro points) and cut single-query latency (2 threads) from 18 to 6 ms for v2-small and from 48 to 32 ms for v2-base. Dynamic int8 with per-tensor scales is faster still (4 and 7 ms) but *not acceptable* here: −2.44 macro points for v2-small and −5.42 for v2-base (−8.27 on ar→en for the base model). Per-channel scales reduce the loss to −0.72 and −4.56, still more than we would ship for the base model. Weight-only 8-bit quantisation is lossless within noise (+0.04 and −0.11) and shrinks the models from 448 to 114 MiB and from 1059 to 270 MiB, but its CPU kernels are slower per query than fp32 (18 and 97 ms). We therefore recommend ONNX fp32 when latency matters, weight-only 8-bit when download size matters, and no dynamic int8 for these models without calibration or quantisation-aware training.

7 Limitations

Most fine-tuned numbers are single seeds; seeds are complete only for the raw v2-m3, and the WiSE blends and the students of other seeds, as well as the final student recipe, are pending (Table 6). The test set is Wikipedia-style extractive QA with 1,190 parallel questions per XQuAD direction, so the intervals are wide for the cross-lingual slices; domain shift to dialects, informal text or long documents is untested. Training queries are machine-translated and not human-verified; the style check shows a constant offset on evaluation questions but cannot rule out subtler translationese. The guard analysis shows a real regression on MIRACL and Mr.TyDi, and the clean subset is small. The MTEB(ara, v1) scores for the v2 models are pending, and only 10 models have been scored on the benchmark. Deployment latency comes from one machine type, with other jobs running, and does not include serving overhead. No RAG-level evaluation is done here.

8 Reproducibility

Every number in this report is generated by `report/make_tables.py` from the committed result files in `results/retrieval_v2`, `results/mteb_ara` and `deploy/results`; run `python report/make_tables.py && cd report && pdflatex main && bibtex main && pdflatex main && pdflatex main`. Training used a single GB10 GPU; the code of the v1 pipeline and the evaluation harness are in the accompanying repository, and the deployment scripts are in `deploy/`. The data are public (licences above); the v2 model weights are [TODO: published after review: Hugging Face repository names].

Model	Task	Type	Retr.	Rerank	STS	Cls.	ar	ar-en
bge-m3	72.0	70.9	73.5	79.4	66.3	64.5	70.4	73.3
multilingual-e5-large	71.4	70.3	74.0	79.6	68.1	61.8	69.5	75.3
multilingual-e5-base	67.9	67.3	68.8	77.1	64.9	59.6	66.5	69.1
multilingual-e5-small	65.6	65.7	66.0	79.2	60.1	58.3	65.0	63.9
arabic-english-e5-small-retrieval	65.0	64.4	66.3	65.9	57.4	59.0	63.1	68.1
paraphrase-multilingual-MiniLM-L12-v2	58.7	62.7	45.2	70.0	63.1	52.0	54.6	66.5
silma-embeddding-matryoshka-v0.1	60.3	61.9	47.1	71.4	59.2	63.2	62.2	47.9
LaBSE	58.2	58.7	46.2	51.6	62.5	58.3	54.2	66.2
Arabic-Retrieval-v1.0	58.3	56.8	55.8	65.2	62.8	64.0	67.4	27.9
Arabic-Triplet-Matryoshka-V2	55.7	54.8	51.0	64.5	60.3	63.4	66.3	19.5
v2-small	[TODO: pending]							
v2-base	[TODO: pending]							
v2-m3	[TODO: pending]							

Table 7: MTEB(ara, v1), scores $\times 100$. Task = mean over tasks, Type = mean over task types; ar / ar-en = mean over Arabic-only / Arabic–English subsets.

Model	MIRACL-HN	Mr.TyDi rerank	SemRel24
mE5-small	72.4	79.2	51.0
v1	65.8	65.8	41.2
mE5-base	73.0	77.2	59.2
bge-m3	79.0	79.4	59.1
v2-m3 (WiSE 0.75)	75.7	72.1	61.8
v2-m3 − bge-m3	−3.3	−7.3	+2.7

Table 8: Guard tasks (MTEB protocol, $\times 100$): MIRACL-HardNegatives (nDCG@10), Mr.TyDi reranking (MAP), SemRel24 (Spearman). These runs use slightly different numerics than Table 7 (differences up to 0.5 points for the same model and task).

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Model	MIRACL-HN (n=184+816) all	clean	contam.	Mr.TyDi rerank (n=894+24) all	clean	contam.
bge-m3	75.7	73.1	76.2	79.4	79.1	89.2
mE5-base	70.4	67.3	71.1	77.4	77.2	82.6
v1	64.0	61.8	64.5	65.8	65.6	72.5
v2-m3 raw FT	66.8	64.9	67.3	69.4	69.3	73.8
v2-m3 (WiSE 0.75)	70.2	68.8	70.6	72.1	71.9	79.3

Table 9: Guard tasks split by contamination: "clean" queries are not near-duplicates of any TyDi-ar or MIRACL-ar training question, "contam." queries are. Per-query scores in the contamination-split protocol.

Variant (CPU)	MiB	RSS	q@1 ms	p/s	MRR	Δ	Δ ar→en
v2-small PyTorch fp32	449	1818	18	14	77.4	ref.	ref.
v2-small ONNX fp32	448	2613	6	20	77.4	+0.00	+0.00
v2-small ONNX int8	113	2249	4	34	75.0	−2.44	−5.09
v2-small ONNX int8, per-channel	113	2258	4	33	76.7	−0.72	−1.68
v2-small ONNX weight-only 8-bit (MatMulNBits)	114	2266	18	20	77.5	+0.04	+0.05
v2-base PyTorch fp32	1061	2447	48	3	79.7	ref.	ref.
v2-base ONNX fp32	1059	3089	32	4	79.7	+0.00	+0.00
v2-base ONNX int8	265	2708	7	11	74.3	−5.42	−8.27
v2-base ONNX int8, per-channel	266	2736	9	11	75.1	−4.56	−6.50
v2-base ONNX weight-only 8-bit (MatMulNBits)	270	2226	97	4	79.6	−0.11	−0.18

Table 10: CPU deployment variants. MiB: size on disk; RSS: peak resident memory of the benchmark process (8 threads; includes the python and data-loading baseline of that process); q@1: p50 latency of a single query (2 threads, ms); p/s: passages per second (batch 32, 8 threads); MRR: test macro MRR@10 ×100; Δ: difference to PyTorch fp32 in points (macro, and ar→en).

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